

SERAT MAHMUD SAAD

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RESEARCH INTERESTS

Observational Astronomy: Stellar Astrophysics, AGN and Quasars, Exoplanets

Computational Techniques: Machine Learning, AI, Quantum Computing

EDUCATION

Ph.D. in Astrophysics (in progress), The Ohio State University 2025-Present
Advisor: Prof. Yuan-Sen Ting

A.B. in Physics & Applied Mathematics, Vanderbilt University 2021-2025
Thesis: Understanding Accretion Activity in Young Stars using SDSS-V BOSS Spectra
Advisor: Prof. Keivan Stassun

HONORS & AWARDS

University Distinguished Fellowship, The Ohio State University	2025-2026
University Fellowship, The Ohio State University	2025
Larry R. Cathey Award, Vanderbilt University	2025
Highest Honors in Astronomy, Vanderbilt University	2025
International Full-ride Student Grant, Vanderbilt University	2021-2025
Dean's List, Vanderbilt University	2021-2025
Pi Mu Epsilon, Vanderbilt University	2024
Sigma Pi Sigma, Vanderbilt University	2023
Buchanan Fellowship, Vanderbilt University	2022
Silver Medal, International Olympiad on Astronomy and Astrophysics	2020

FIRST AUTHOR PUBLICATIONS

1. **Serat Saad**, Marina Kounkel, Keivan G. Stassun et al. *ABYSS III: Observing accretion activity in young stars through empirical veiling measurements*. June, 2025. The Astronomical Journal. Doi: 10.3847/1538-3881/ade43b
2. **Serat Saad**, Kaitlyn Lane, Marina Kounkel, Keivan G. Stassun et al. *ABYSS II: Identification of young stars in optical SDSS spectra and their properties*. January, 2024. The Astronomical Journal. Doi: 10.3847/1538-3881/ad2001
3. **Serat Saad**, Jessie C. Runnoe, Michael Eracleous et al. *Reconsidering Tick Tock: An alternative to the supermassive black hole binary hypothesis of J1430*. Will submit to ApJ.

CO-AUTHOR PUBLICATIONS

1. Jessie C. Runnoe, **Serat Saad**, Michael Eracleous, Niana Mohammed et al. *Is the supermassive black hole binary candidate J0950+5128 actually a single perturbed accretion disk?*. Will submit to ApJ.
2. Mojgan Aghakhanloo, et al. including **Serat Saad**. *The Nineteenth Data Release of the Sloan Digital Sky Surveys*. Will submit to ApJS.

TELESCOPE TIMING PROPOSALS

1. *Principal Investigator*, Large Binocular Telescope (PEPSI), 2025B. "PASTA Survey: High precision stellar chemical abundance for 43 planet-hosting stars."
2. *Co-Investigator*, Chandra X-ray Observatory, Cycle 26, 2024. General Observing. "Pink Mergers: Dwarf Galaxies with Extremely Luminous Quasars." Principal Investigator: Prof. Adi Foord..

CONFERENCE POSTERS AND ABSTRACTS

1. **Serat Saad**, Jessie C. Runnoe, Michael Eracleous. *An Alternative to the Supermassive Black Hole Binary Hypothesis for J1430 (Tick Tock)*. AAS Winter Conference 2025. Doi: 2025aas/24535502S
2. **Serat Saad**, Marina Kounkel, Keivan G. Stassun. *Using low resolution optical spectra to identify young stars and their properties*. Cambridge Cool Stars Conference 2024. Doi: 10.5281/zenodo.13007729
3. **Serat Saad**, Anish Giri, Mubarak Ganiyu, David Nizovsky et al. *Developing Workforce in Quantum Industry: The Wond'ry Qunatum Studio*. IEEE International Conference on Quantum Computing and Engineering (QCE) 2023. Doi: 10.1109/QCE57702.2023.10282

TEACHING EXPERIENCES

TA, DS 2100: Statistics for data science., Vanderbilt University	Spring 2025
TA, ASTR 1010: Introductory Astronomy, Vanderbilt University	Spring 2024
Academic Mentor, Bangladesh Olympiad on Astronomy and Astrophysics	2021-2024
Grader, DS 1100: Introduction to data science, Vanderbilt University	Fall 2023
Instructor, Quantum Computing for everyone, TWQS Vanderbilt	Fall 2023
Grader, CS 1100: Introduction to problem solving with python.	2022-2023
Team Leader, International Olympiad on Astronomy and Astrophysics	2022
Tutor, Vanderbilt Student Athletics	2022

PROFESSIONAL ORGANIZATIONS

American Astronomical Society (AAS)	2024-Present
Milky Way Mapper (MWM) program, Sloan Digital Sky Survey (SDSS)	2021-Present
Institute of Electrical and Electronics Engineers (IEEE)	2022-2024